

Emile Gaudinot

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EDUCATION

- **Technical University of Berlin** Berlin, Germany
Applied Mathematics, Statistics, Data Science (MSc), GPA: 4.0 2022 – 2023
- **École Centrale de Nantes** Nantes, France
Engineering Degree (MSc), GPA: 4.0 2020 – 2022

EXPERIENCE

- **Sia** Paris, France
Experienced Data Science Consultant Feb. 2026 – Present (9 months)
 - **Revenue Forecast (Luxury and Retail Client)**: Trained and sent to production a **Holt-Winters** model on **GCP Vertex AI**. Predicted sales over 12 months, with a daily granularity, for each of the 600 stores in the world. **-17%** WMAPE as compared to the sales plans.
 - **Electrical Network RL Optimization (Major Energy Provider)**: Using **Reinforcement Learning** and **PPO**, optimized the (non-)topological actions to avoid network congestion. **+500%** speed as compared to current solvers.
 - **Seine River Flood Forecasting (Public Service Client)**: Forecasted water levels under Austerlitz bridge using **open-source APIs** Hub'eau, ISD, Météo-France, **ERA5**. Developed an **XGBoost** model with **SHAP values** for explainability. **+24%** accuracy compared to *vigicrues.gouv.fr*.
 - **Video Moderation System (Media Client)**: Automated the blurring of alcohol, cigarettes and brands with a **YOLO26-large** model, on **AWS SageMaker**. Agentic pipeline with **Langflow** to ensure videos respect compliance.
- **Fraunhofer - Heinrich-Herz-Institute** Berlin, Germany
Experienced Data Scientist Mar. 2025 – Feb. 2026 (1 year)
 - **Perceived Distance Prediction from Brain Signals**: Developed an innovative AI pipeline using **MNE-Python** with (1) an **SSD** and a **SPoC regression** for distance, and (2) a **ROC classifier** for personal space intrusion detection ($F1 = 0.81$). Extension of the pipeline for **real-time** using **LSL** ($F1 = 0.65$) and deployed it with **Databricks Workflows**. Published the results in a [research paper](#).
- **Charité** Berlin, Germany
Data Scientist Mar. 2023 – Feb. 2025 (2 years)
 - **Computer Vision for Early Detection of Coronary Artery Disease**: Segmented arteries from 3D heart images with **U-Net**, **U-net Transformer**, and **MedSAM**, on **MONAI** ($IoU = 0.79$). Implemented the pipeline for artery graph extraction with **scikit-image** (**CCA**, **skeletonization**, etc). Monitoring with **MLflow**. Release in production with **Databricks**, leading to **+60%** correct CAD diagnosis. Published the results in a [research paper](#).
 - **Advanced ML Solutions for Computer Security**: Implemented a malware detector for .pdf and .docx using **Bag-of-Words**, **PCA** and **XGBoost** ($F1 = 0.98$). Automated the network intrusion detector using **Isolation Forests** and **LOF scores** (**-120%** false positives). Robustified models to **Adversarial Attacks** using **i-FGSM**.

PROJECTS

- **Winner of BIFOLD's Cybersecurity Hackathon (Oct. 2023 – Mar. 2024)**: Trained and fine-tuned ML models on 4 cybersecurity challenges. Price Ceremony (1st Price/24) at the German Ministry of Research, Technology and Space.
- **Local LLM - Complete app (Apr. 2025 – Nov. 2025)**: Developed and optimized a custom [ChatBot application](#), which runs locally. Possibility to upload documents (img, pdf, docx, pptx) thanks to an OCR + text extraction pipeline.

SKILLS

- **LLMs**: Mistral API, Ollama, Light local models (mistral-7b, gemma4-12b, llama3.1-8b)
- **Deep Learning**: Time series (RNN, LSTM), Segmentation (nnU-Net, UNETR, MedSAM), Object detection (YOLO26)
- **Tools**: Production (AWS, GCP, Databricks, Docker, MLflow, Git), Python (PyTorch, Scikit-learn, Pandas), HuggingFace
- **Programming Languages**: Python, SQL, Bash, C++, R
- **Spoken Languages**: French (native), English (C1 - TOEFL 104/120), German (C1 - ZEMS 92/100)

Certifications: Databricks Certified Machine Learning Associate, ANSSI SecNum: Cybersecurity & Digital Tool Protection